



WEEK 2 · UNIT: ENGINEERING MATERIALS

Ferrous Engineering Materials

Classification of materials · Cast Iron · Steel · Alloy Steel

What We'll Cover This Week

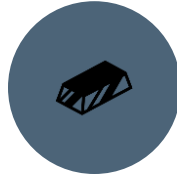
01



Classifying Materials

How engineers group all materials into ferrous, non-ferrous and non-metals.

02



Cast Iron

Grey, White & Malleable cast iron — composition, properties and uses.

03



Steel Basics

How steel is classified and named (designation) by its carbon content.

04

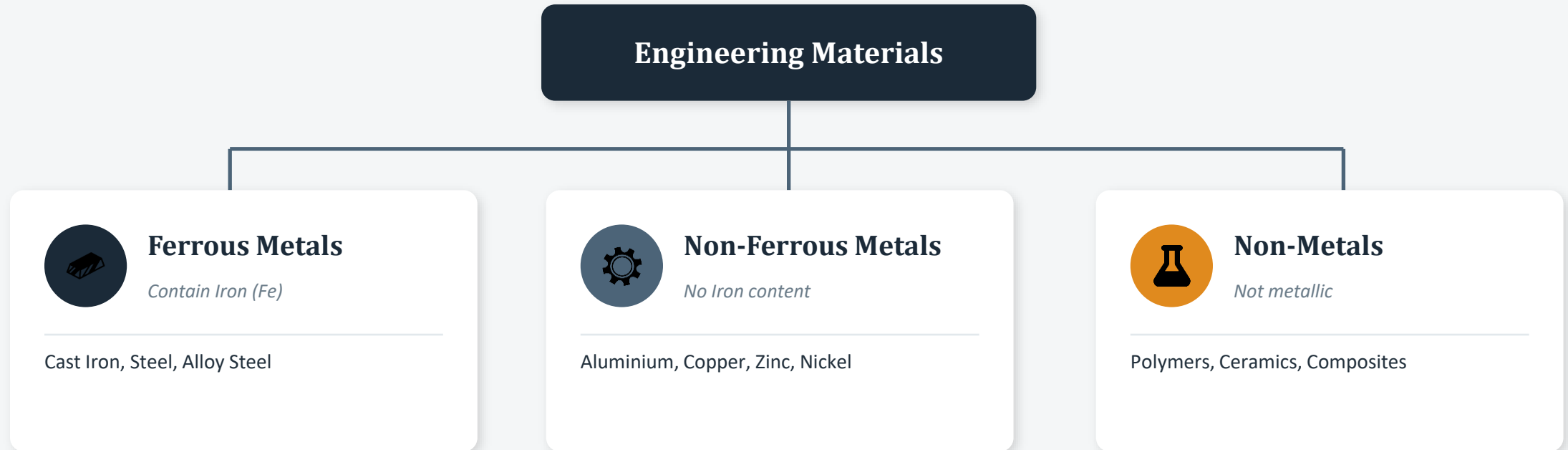


Alloy Steels

Tool steel, tungsten steel, stainless steel & high-speed steel.

Classification of Engineering Materials

Every material an engineer selects belongs to one of these families:



Today's focus: Ferrous Metals — materials that contain iron as the main element.

FERROUS METALS

Ferrous metals are metals that contain iron as the main element.

WROUGHT IRON		CAST IRON		
				
Wrought Iron	Grey Cast Iron	White Cast Iron	Malleable Cast Iron	Ductile (Nodular) Cast Iron

STEEL				
LOW CARBON STEEL	MEDIUM CARBON STEEL	HIGH CARBON STEEL	ALLOY STEEL	STAINLESS STEEL
				
Low Carbon Steel	Medium Carbon Steel	High Carbon Steel	Alloy Steel	Stainless Steel

KEY POINTS

- Ferrous metals contain iron and are generally magnetic.
- They have good strength and are widely used in construction, machinery, automotive, and tools.
- Examples: Wrought Iron, Cast Iron, and Steel (various types).

What is Cast Iron?

Simple definition

Cast iron is an iron alloy containing **more than 2% carbon** that can be easily **melted and poured into molds to make machine parts**.

It contains **more carbon than steel**, which makes it easy to melt and cast into complex shapes, but more brittle.

Typical Composition

- Iron (Fe) **~ 93 - 95 %** — *base metal*
- Carbon (C) **2 - 4 %** — *gives hardness, castability*
- Silicon (Si) **1 - 3 %** — *helps carbon form graphite*
- Manganese, Sulphur, Phosphorus **small %** — *trace elements*

Key idea: More carbon than steel → easy to cast, but brittle (breaks instead of bending).



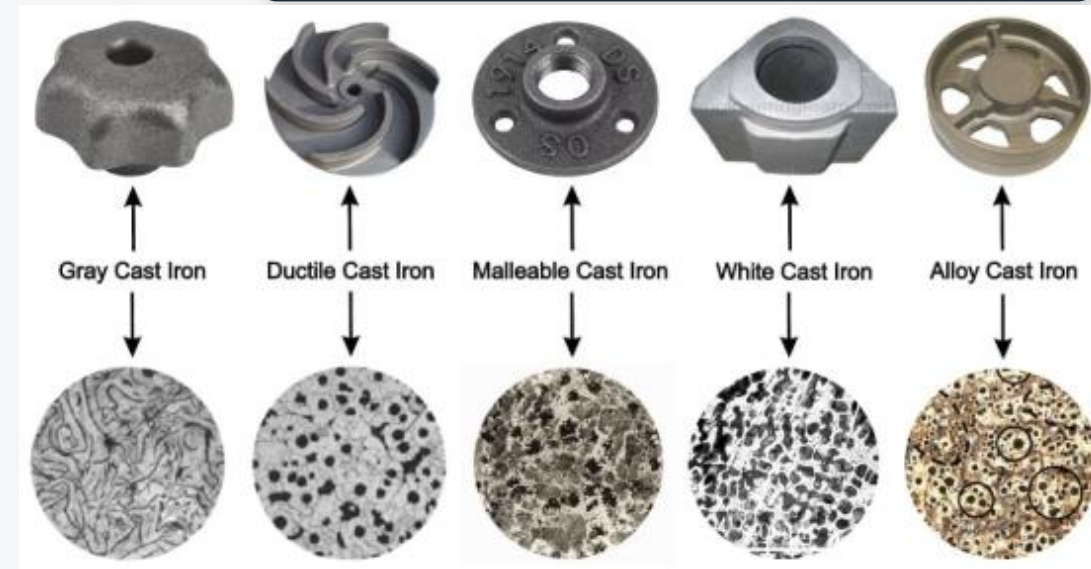
3 Main Types of Cast Iron

Grey Cast Iron

White Cast Iron

Malleable Cast Iron

→ details on the next slide



Grey Cast Iron: Properties

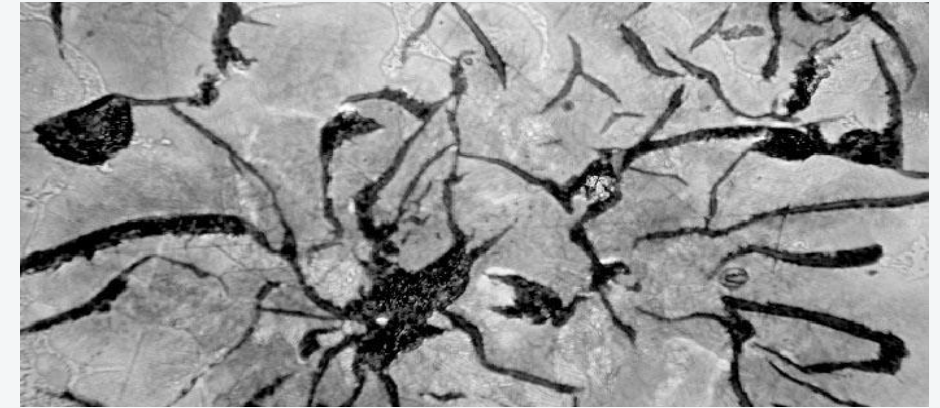


Grey Cast Iron

Grey cast iron is a common engineering material used in **machine beds, engine blocks, pipes, and flywheels**. Its broken surface looks grey because **the carbon is present as graphite flakes**.

Main properties

Property	Simple meaning
Good machinability	Easy to cut, drill, and shape.
Good wear resistance	Resists rubbing and surface damage.
High compressive strength	Can withstand heavy pressing loads.
Good vibration damping	Absorbs vibration and noise well.
Good castability	Molten metal flows easily into moulds.
Brittle	Breaks suddenly under impact.
Poor tensile strength	Not very strong when pulled.
Not easily forged	Cannot be hammered into shape.



Grey Cast Iron: Properties & Uses



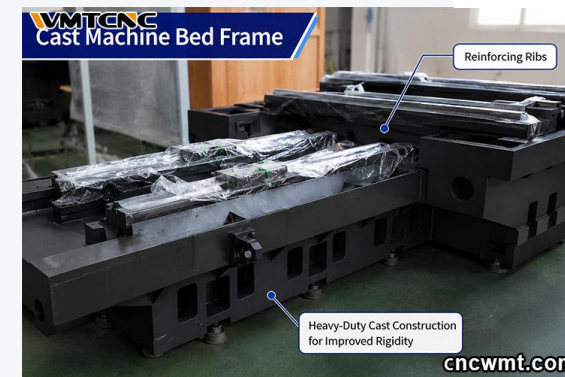
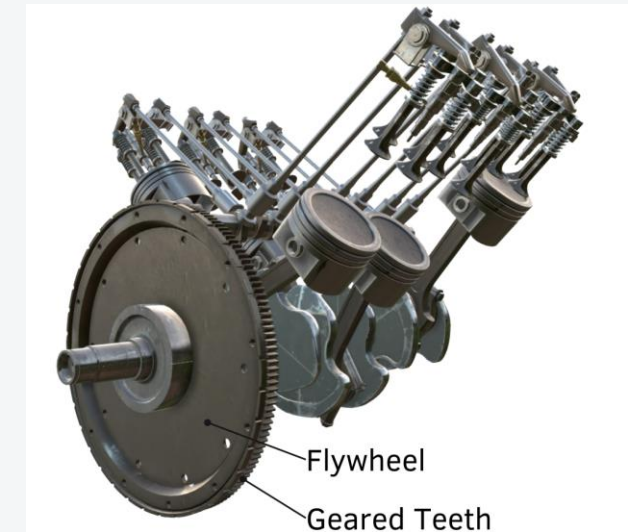
Grey Cast Iron

Composition

Element	Percentage (%)	Purpose
Carbon (C)	2.5–4.0	Provides hardness and forms graphite flakes.
Silicon (Si)	1.0–3.0	Promotes graphite formation and improves castability.
Manganese (Mn)	0.2–1.0	Increases strength and hardness.
Phosphorus (P)	0.05–1.0	Improves the flow of molten metal during casting.
Sulphur (S)	0.02–0.20	Kept low to avoid brittleness.
Iron (Fe)	Balance	Main metal.

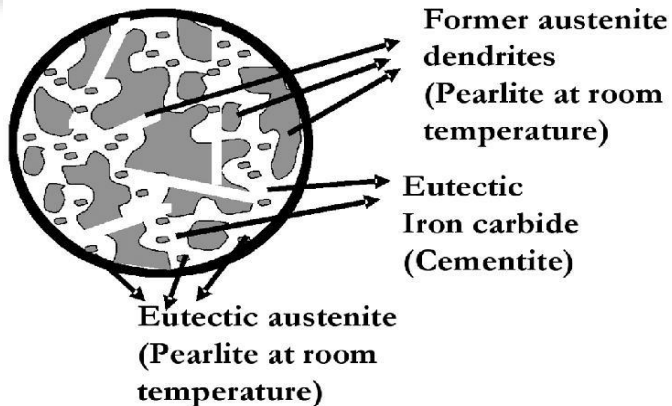
APPLICATIONS

- *Machine tool beds,*
- *flywheels,*
- *pipes,*
- *manhole covers*





White Cast Iron

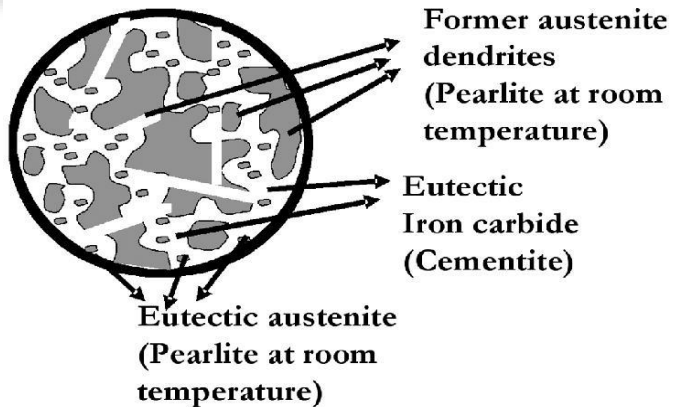


1. White cast iron is a type of cast iron that is **very hard and wear-resistant**, but it is also **brittle** (breaks easily if hit hard).
2. It gets its name because its **broken surface looks white and shiny** due to the presence of iron carbide (cementite).

Property	Simple Meaning
Very hard	Resists scratching and wear.
Excellent wear resistance	Lasts a long time in rubbing or abrasive conditions.
High compressive strength	Can withstand heavy pressing loads.
Very brittle	Breaks easily if hit or dropped.
Poor machinability	Difficult to cut, drill, or machine.
Poor weldability	Difficult to weld without cracking.
Low ductility	Cannot bend or stretch much before breaking.
Not forgeable	Cannot be shaped by hammering.



White Cast Iron



White Cast Iron – Composition

Element	Percentage (%)	Purpose
Carbon (C)	1.8–3.6	Increases hardness and wear resistance. Carbon is mainly present as iron carbide (cementite).
Silicon (Si)	0.5–1.9	Kept low to prevent graphite formation.
Manganese (Mn)	0.2–1.0	Increases hardness and strength.
Phosphorus (P)	Less than 0.20	Improves fluidity during casting but is kept low.
Sulphur (S)	Less than 0.12	Kept low to reduce brittleness.
Iron (Fe)	Balance	Main metal.

APPLICATIONS OF WHITE CAST IRON



1. GRINDING BALLS

Used in ball mills to grind and crush ores and minerals.



2. CRUSHER JAWS

Used in stone and ore crushers to crush large rocks.



3. MILL LINERS

Protect the inside walls of grinding mills from wear.



4. PUMP IMPELLERS

Used in pumps that handle abrasive slurries and tough fluids.



5. ROLLERS

Used in rolling mills, sugar mills and crushing equipment.



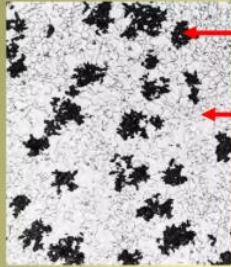
6. WEAR PLATES

Used as wear-resistant surface where there is high abrasion.



Malleable Cast Iron

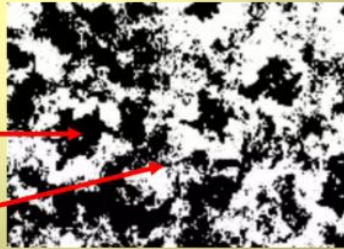
Microstructure of malleable cast iron



Graphite

Ferrite

Ferritic Malleable Cast Iron



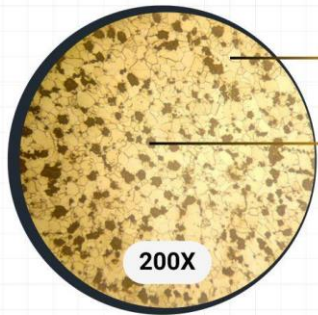
Graphite

Pearlite

Cast Iron

//Metallography

ETCHED MALLEABLE CAST IRON



Ferrite

Tempered Carbon

200X



Microstructural features: **Tempered carbon** (Dark region) and **Ferrite** (White regions)



Etchant: **Nital**



@metallurgydaily

Save

Malleable cast iron is produced by **heat treating white cast iron**. During heat treatment, the **hard cementite** changes into **temper carbon (small rounded graphite particles)**, making the **material tougher and less brittle**

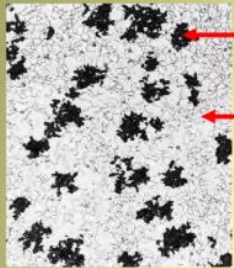
Property	Simple Meaning
Good ductility	Can bend without breaking.
High toughness	Can absorb shocks and impacts.
Good machinability	Easy to cut, drill, and machine.
Good tensile strength	Can withstand pulling forces.
Good wear resistance	Resists surface wear during use.
Good impact resistance	Does not crack easily when struck.
Good weldability	Easier to weld than white cast iron.
Less brittle	Stronger and more flexible than white cast iron.



Malleable Cast Iron

Malleable cast iron is cast iron that becomes stronger and less brittle after heat treatment, so it can bend slightly without breaking.

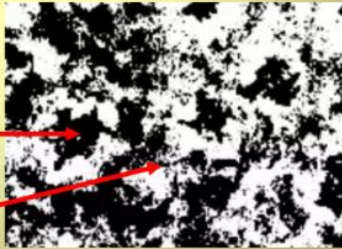
Microstructure of malleable cast iron



Graphite

Ferrite

Ferritic Malleable Cast Iron



Graphite

Pearlite

Pearlitic Malleable Cast Iron

Composition of Malleable Cast Iron

Element	Percentage (%)	Purpose
Carbon (C)	2.0–2.6	Gives strength and hardness
Silicon (Si)	0.9–1.9	Helps graphite formation
Manganese (Mn)	0.2–1.0	Increases strength
Sulphur (S)	Less than 0.18	Kept low to avoid brittleness
Phosphorus (P)	Less than 0.20	Improves fluidity but kept low
Iron (Fe)	Balance	Main metal

Applications

1. Pipe fittings
2. Railway Components
3. Agricultural components
4. Hand Tools



Comparison — Grey vs White vs Malleable

Property	Grey CI	White CI	Malleable CI
Carbon form	<i>Graphite flakes</i>	<i>Cementite</i>	<i>Graphite nodules</i>
Hardness	<i>Moderate</i>	<i>Very high</i>	<i>Moderate</i>
Machinability	<i>Good</i>	<i>Poor</i>	<i>Good</i>
Strength	<i>Moderate</i>	<i>High (brittle)</i>	<i>Good, tough</i>
Typical use	<i>Machine beds</i>	<i>Wear plates</i>	<i>Pipe fittings</i>





DEFINITION OF STEEL

Steel is an alloy of iron and carbon. It contains 0.02% to 2.1% carbon, with small amounts of other elements. It is strong, tough and widely used in construction, machines, tools, vehicles and many other applications.



The amount of **carbon** in steel affects its **strength**, **hardness** and **ductility**.

TYPE OF STEEL	CARBON CONTENT	KEY FEATURES	COMMON USES
1. LOW CARBON STEEL (MILD STEEL) 	Up to 0.25% Carbon (C) 	• Soft and ductile • Easy to weld and form • Low strength • Good machinability	 • Building structures • Pipes • Automobile bodies • General fabrication
2. MEDIUM CARBON STEEL 	0.25% – 0.60% Carbon (C) 	• Higher strength than mild steel • Moderate ductility • Can be heat treated	 • Machine parts • Shafts • Gears • Railways components
3. HIGH CARBON STEEL 	0.60% – 1.50% Carbon (C) 	• Very high strength • Hard but less ductile • Wear and heat resistant • Can be hardened and tempered	 • Springs • Cutting tools • Knives • High strength wires



NOTE : As carbon content increases → **hardness and strength increase**, but **ductility decreases**.

STEEL DESIGNATION (NAMING OF STEEL)



Steel is given a name or code (designation) to show its type, composition, and properties.

1. IS SYSTEM (INDIAN STANDARD)

- Steel is designated by numbers.
- The number indicates the **minimum tensile strength** in N/mm² (MPa).

EXAMPLE

IS 2062 → Minimum tensile strength of **410 N/mm²**.



Common IS Steel Designations

Designation	Minimum Tensile Strength (N/mm ²)	Typical Use
IS 2062 Gr. E250	250	General structures
IS 2062 Gr. E350	350	Bridges, buildings
IS 2062 Gr. E410	410	Heavy structures
IS 2062 Gr. E550	550	High rise buildings

How to Read an IS Designation?



2. AISI / SAE SYSTEM (AMERICAN SYSTEM)

- Steel is designated by 4 digits.
- First two digits → Type of steel (10xx – Carbon steel, 41xx – Alloy steel, etc.)
- Last two digits → Approximate carbon content in hundredths of a percent.

EXAMPLE

1045 →
10 → Carbon steel
45 → 0.45% Carbon



Common AISI / SAE Designations

AISI / SAE No.	Type	Carbon Content (Approx.)	Typical Use
1018	Carbon Steel	0.18%	Shafts, axles
1045	Carbon Steel	0.45%	Machine parts
4140	Alloy Steel (Cr-Mo)	0.40%	Gears, bolts, shafts
4340	Alloy Steel (Ni-Cr-Mo)	0.40%	High strength parts

3. EN SYSTEM (EUROPEAN SYSTEM)

- Steel is designated by EN numbers.
- It indicates the type of steel and its properties.

EXAMPLE

EN Designation	Description
EN 8	Medium carbon steel
EN 24	Alloy steel (Ni-Cr-Mo)
EN 31	Nickel steel
EN 42	Spring steel



In simple words: Steel designation is a code that tells us the **type of steel** and its **strength or use**.

LOW CARBON STEEL (MILD STEEL)

1. DEFINITION

Low carbon steel (mild steel) is a type of steel that contains a small amount of carbon. It is soft, ductile, easy to weld and form, and widely used in construction and manufacturing.



2. COMPOSITION (Typical)

Element	C	Mn	P (Max)	S (Max)	Si	Fe
Percentage (Approx.)	Up to 0.25%	0.30 – 0.70%	0.04%	0.05%	0.10 – 0.40%	Balance

Key point: Carbon content is **up to 0.25%**.

3. PROPERTIES

- Soft and ductile
- Good weldability
- Easy to form and machine
- Low to moderate strength
- Good toughness
- Good fatigue resistance
- Economical and widely available

SUMMARY



Low carbon steel contains up to 0.25% carbon. It is soft, ductile, easy to weld and form. It is used in construction, pipes, automotive parts, furniture and general engineering applications.

TYPICAL PRODUCTS



Bars & Rods



Plates & Sheets



Pipes



Structural Sections

4. APPLICATIONS



Building structures



Pipes & Tubes



Automobile bodies



Bolts, Nuts & Screws



Furniture



Appliance parts

MEDIUM CARBON STEEL

1. DEFINITION

Medium carbon steel is a type of steel that contains moderate amount of carbon. It has higher strength than low carbon steel, good toughness and can be heat treated. It is widely used in machine parts and structural components.



2. COMPOSITION (Typical)

Element	C	Mn	P (Max)	S (Max)	Si	Fe
Percentage (Approx.)	0.25 – 0.60%	0.60 – 1.20%	0.04%	0.05%	0.15 – 0.40%	Balance

Key point: Carbon content is **0.25% – 0.60%**.

3. PROPERTIES

- Higher strength than low carbon steel
- Good toughness
- Moderate ductility
- Can be heat treated (hardened and tempered)
- Good wear resistance
- Good machinability
- Can withstand heavy loads and impact

SUMMARY



Medium carbon steel contains 0.25% – 0.60% carbon. It has higher strength and toughness than low carbon steel, can be heat treated and is used in gears, shafts, machine parts, automobile components and structural parts.

TYPICAL PRODUCTS



Gears & Shafts



Crankshafts



Connecting Rods



Machine Components

4. APPLICATIONS



Machine parts



Railway components



Bolts & Fasteners



Automobile parts



Agricultural implements

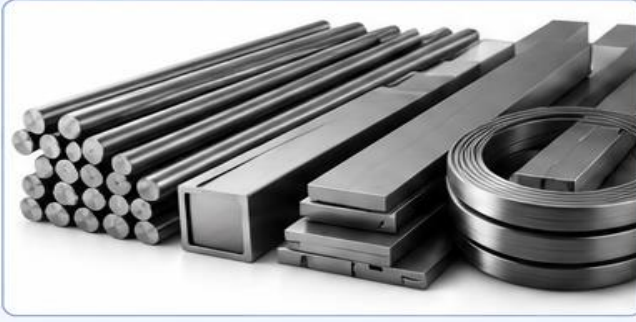


Heavy machinery parts

HIGH CARBON STEEL

1. DEFINITION

High carbon steel is a type of steel that contains a high amount of carbon. It is very hard and strong but less ductile. It can be heat treated (hardened and tempered) to get high strength and wear resistance. It is used where high strength and hardness are required.



2. COMPOSITION (Typical)

Element	C	Mn	Si	P (Max)	S (Max)	Fe
Percentage (Approx.)	0.60 – 1.50%	0.30 – 0.60%	0.15 – 0.35%	0.04%	0.05%	Balance

Key point: Carbon content is **0.60% – 1.50%**.

3. PROPERTIES

- Very high hardness and strength
- Excellent wear resistance
- Can be hardened and tempered
- High tensile strength
- Low ductility
- Brittle in nature
- Poor weldability
- Lower toughness compared to low and medium carbon steel

SUMMARY



High carbon steel contains 0.60% – 1.50% carbon. It is very hard, strong and wear resistant but less ductile. It is heat treated to increase hardness and strength and is used in cutting tools, springs, wires, knives and other high strength applications.

TYPICAL PRODUCTS



Cutting Tools



Drills & Reamers



Springs



Knives & Blades



Files



High Strength Wires

4. APPLICATIONS



Cutting Tools



Springs



Drills & Dies



Knives & Blades



High Strength Wires



Files & Hand Tools

ALLOY STEEL

1. DEFINITION

Alloy steel is steel that contains **other elements** (alloying elements) in addition to iron and carbon.

These elements are added to **improve** its strength, hardness, toughness, wear resistance and other **properties** for specific applications.



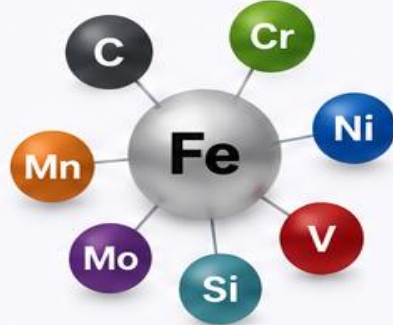
2. KEY POINTS

- ✓ Contains elements like **Cr, Ni, Mn, Mo, Si, V**, etc.
- ✓ Better **strength, hardness** and **toughness** than carbon steel.
- ✓ Resistant to **wear, corrosion** and **heat**.
- ✓ Can be **heat treated** to get desired properties.
- ✓ Used in special applications where high performance is required.



3. COMPOSITION (Typical)

- **Iron (Fe)** – Base metal
- **Carbon (C)** – 0.10% to 1.50% (varies)
- **Alloying Elements** – Added in small amounts to improve properties



4. APPLICATIONS



Gears & Shafts



Automobile Parts



Aerospace Components



Oil & Gas Equipment



Construction & Heavy Machinery

5. COMMON ALLOYING ELEMENTS & THEIR EFFECTS

Chromium (Cr)



Increases hardness, wear and corrosion resistance.

Nickel (Ni)



Improves toughness, strength and corrosion resistance.

Manganese (Mn)



Increases strength, hardness and wear resistance.

Molybdenum (Mo)



Improves strength at high temperature and resists softening.

Silicon (Si)



Increases strength and improves oxidation resistance.

Vanadium (V)



Refines grain structure and increases strength and wear resistance.

NOTE



By adding different alloying elements in proper amounts, alloy steels can be made suitable for a wide range of demanding applications.

TOOL STEEL

DEFINITION

Tool steel is a special type of steel that is used to make tools. It is very **hard**, **wear resistant** and can keep its shape even at **high temperature**.



1. COMPOSITION (Typical)

Element	Percentage (Approx.)
Carbon (C)	0.50 – 1.50%
Chromium (Cr)	1.00 – 12.00%
Tungsten (W)	0.50 – 18.00%
Molybdenum (Mo)	0.10 – 5.00%
Vanadium (V)	0.10 – 2.00%
Manganese (Mn)	0.10 – 1.00%
Silicon (Si)	0.10 – 1.00%
Iron (Fe)	Balance



Exact composition varies depending on the type and use of tool steel.

2. PROPERTIES

- ✓ Very high hardness
- ✓ Excellent wear resistance
- ✓ High strength and toughness
- ✓ Can withstand high temperature
- ✓ Holds cutting edge (does not dull easily)
- ✓ Can be heat treated to get desired hardness and strength
- ✓ Long life and dimensional stability



3. APPLICATIONS



Cutting tools

Drills, taps, reamers, milling cutters



Forming tools

Dies, punches, moulds



Moulds

Plastic moulds, die casting moulds



Measuring tools

Gauges, templates, micrometers



Press tools

Used in pressing, blanking, piercing operations



Other tools

Saws, knives, chisels, etc.



KEY POINT: Tool steel is designed to be harder and stronger than ordinary steel so that it can cut, shape or form other materials and last for a long time.



**Harder Tools
Better Performance
Longer Life**

TUNGSTEN STEEL (HIGH SPEED STEEL)

DEFINITION

Tungsten steel (High Speed Steel – HSS) is a type of tool steel that contains tungsten as the main alloying element.

It can **stay hard even at high temperature**, has **excellent wear resistance** and is used for high speed cutting tools.



1. COMPOSITION (Typical)

Element	Percentage (Approx.)
Tungsten (W)	6.00 – 18.00%
Carbon (C)	0.70 – 1.40%
Chromium (Cr)	3.00 – 4.50%
Vanadium (V)	1.00 – 3.00%
Molybdenum (Mo)	4.00 – 8.00%
Cobalt (Co)	0.00 – 10.00%
Silicon (Si)	0.20 – 1.00%
Manganese (Mn)	0.10 – 0.50%
Iron (Fe)	Balance



Tungsten is the key element which gives high red hardness (hardness at high temperature) and wear resistance.

2. PROPERTIES

- ✓ Very high hardness (even at high temperature)
- ✓ Excellent wear resistance
- ✓ High red hardness (retains hardness up to **600–650°C**)
- ✓ High compressive strength
- ✓ Good toughness and resistance to softening
- ✓ Can be heat treated to achieve required hardness and performance
- ✓ Long tool life and dimensional stability



Red Hardness
Retains hardness
up to
600 – 650°C

3. APPLICATIONS



Cutting Tools

Drills, taps, reamers, milling cutters, broaches



Metal Cutting

High speed cutting of steel, stainless steel and other metals



Tool Room Work

Used in tool making and precision engineering



Forming Tools

Dies, punches, shear blades



Production Tools

Used in mass production tools requiring high speed and long life



KEY POINT: Tungsten steel (HSS) contains tungsten which allows the steel to keep its hardness even at high temperature, making it ideal for high speed cutting tools.



High Hardness
at High Temp.



Excellent
Wear Resistance



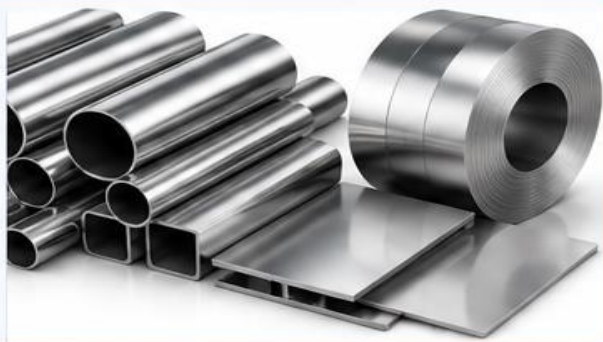
Long
Tool Life

STAINLESS STEEL



DEFINITION

Stainless steel is a special type of steel that contains minimum 10.5% chromium. It is “stainless”, meaning it **does not rust or corrode easily** because a thin, strong protective layer forms on its surface. It is strong, durable, attractive and easy to clean, so it is used in many industries and daily life.



KEY POINTS

- ✓ Contains minimum **10.5% Chromium (Cr)**.
- ✓ Forms a thin protective oxide layer that **prevents rust and corrosion**.
- ✓ Good strength, toughness and ductility.
- ✓ Excellent resistance to corrosion, heat and stains.
- ✓ Hygienic, easy to clean and maintains shine.
- ✓ Can be welded and formed easily.
- ✓ 100% recyclable and eco-friendly.



1. COMPOSITION (Typical)

Element	Percentage (Approx.)
Chromium (Cr)	10.5 – 30%
Nickel (Ni)	0 – 20%
Carbon (C)	0.03 – 0.08%
Manganese (Mn)	0 – 2%
Silicon (Si)	0 – 1.0%
Molybdenum (Mo)	0 – 6%
Iron (Fe)	Balance



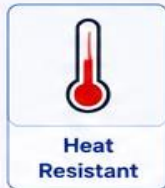
The main element that gives stainless property is **Chromium** (minimum 10.5%). Nickel and Molybdenum improve strength and corrosion resistance.

2. PROPERTIES

- ✓ Excellent corrosion and rust resistance
- ✓ High strength and toughness
- ✓ Good ductility and formability
- ✓ Heat resistant (withstands high temperature)
- ✓ Stain and oxidation resistant
- ✓ Hygienic and non-toxic
- ✓ Low maintenance and long life
- ✓ Attractive shiny surface



Corrosion
Resistant



Heat
Resistant



Easy to Clean
& Hygienic

3. APPLICATIONS



Kitchen & Utensils

Sinks, pans, vessels, cutlery, kitchen equipment



Chemical Industry

Tanks, pipes, valves, heat exchangers



Construction

Railings, doors, windows, roofs, cladding



Medical & Surgical

Surgical instruments, hospital equipment



Automotive

Exhaust systems, trim, decorative parts



Home Appliances

Refrigerators, washing machines, ovens

TYPES (Examples)



- Austenitic Stainless Steel (e.g., 304, 316) – Most widely used, excellent corrosion resistance.
- Ferritic Stainless Steel (e.g., 430) – Good corrosion resistance, magnetic.
- Martensitic Stainless Steel (e.g., 410) – High strength and hardness.
- Duplex Stainless Steel – High strength and corrosion resistance.

SUMMARY



Stainless steel is rust-resistant steel that contains minimum 10.5% chromium. It is strong, durable, corrosion resistant, easy to clean and widely used in many fields.

WHY STAINLESS?



Rust
Resistant



Long
Lasting



Attractive
Finish



Recyclable &
Eco-Friendly

HIGH-SPEED STEEL (HSS)

1. DEFINITION

High-Speed Steel (HSS) is a special type of tool steel that can **cut** materials at **very high speeds** **without losing its hardness**.

It contains **tungsten, molybdenum, chromium, vanadium** and other **elements**.

It is used to make cutting tools that work at high temperature and high speed.



KEY FEATURES

- ✓ Remains hard even at high temperature (**600 – 650 °C**).
- ✓ Excellent wear resistance and toughness.
- ✓ Can cut materials at high speed.
- ✓ Can be heat treated.
- ✓ Long tool life.
- ✓ Resistant to softening.



2. COMPOSITION (Typical)

Element	Percentage (Approx.)
Carbon (C)	0.70 – 1.40%
Tungsten (W)	6.00 – 18.00%
Molybdenum (Mo)	4.00 – 10.00%
Chromium (Cr)	3.00 – 4.50%
Vanadium (V)	1.00 – 3.50%
Cobalt (Co)	0 – 10.00%
Silicon (Si)	0.20 – 1.00%
Manganese (Mn)	0.10 – 0.50%
Iron (Fe)	Balance



Tungsten is the main element that gives high red hardness (hardness at high temperature). Molybdenum and vanadium improve hardness and wear resistance.

3. PROPERTIES

- Very high hardness at high temperature (**600 – 650 °C**).
- Excellent wear resistance.
- High compressive strength.
- Good toughness and shock resistance.
- Can cut materials at high speed.
- Can be heat treated to get desired hardness and performance.
- Long tool life and dimensional stability.



Heat
Resistant



Wear
Resistant



Long
Tool Life



High
Performance

4. APPLICATIONS



Cutting Tools

Drills, taps, reamers, milling cutters, broaches



Metal Cutting

High speed cutting of steel and other metals



Forming Tools

Hobs, gear cutters, threading tools



Tool Room Work

Used in tool making and precision engineering



Press Tools & Dies

Punches, dies, blanking tools, cold work tools



Aerospace & Automotive

Parts requiring high strength and wear resistance

TYPES OF HIGH-SPEED STEEL (Examples)



- M2 Steel – General purpose high speed steel.
- M35 Steel – Contains cobalt, higher red hardness and wear resistance.
- T1 Steel – Tungsten based, good for cutting tough materials.
- M42 Steel – High cobalt content, used for heavy duty cutting.

SUMMARY



High-Speed Steel is designed to cut materials at very high speeds. It remains hard at high temperature, has excellent wear resistance and gives long tool life. It is widely used in cutting and forming tools in many industries.

Why is it Important?

- ★ Enables high speed machining
- ★ Improves productivity
- ★ Reduces tool change frequency
- ★ Saves time and cost